

Detailed SQL Notes with Definitions and Examples

1. What is SQL?

SQL (Structured Query Language) is a standard language used to store, retrieve, manipulate, and manage data in relational databases.

Example:

```
SELECT * FROM Employee;
```

2. Features of SQL

- Easy to learn
- Used for querying data
- Supports insert, update, and delete operations
- Provides security and access control
- Supports transactions

3. Database Terminology

Database: Collection of related data.

Table: Collection of rows and columns.

Row: Single record in a table.

Column: Attribute of data.

Primary Key: Uniquely identifies records.

Foreign Key: Establishes relationships between tables.

4. SQL Command Categories

DDL – Data Definition Language

DML – Data Manipulation Language

DQL – Data Query Language

DCL – Data Control Language

TCL – Transaction Control Language

5. DDL Commands

CREATE:

```
CREATE TABLE Employee(  
  Emp_ID INT,  
  Emp_Name VARCHAR(50),  
  Salary DECIMAL(10,2)  
);
```

ALTER:

```
ALTER TABLE Employee  
ADD Department VARCHAR(30);
```

DROP:

```
DROP TABLE Employee;
```

TRUNCATE:
TRUNCATE TABLE Employee;

RENAME:
RENAME TABLE Employee TO Staff;

6. DML Commands

INSERT:
INSERT INTO Employee VALUES(101,'John',50000);

UPDATE:
UPDATE Employee
SET Salary=60000
WHERE Emp_ID=101;

DELETE:
DELETE FROM Employee
WHERE Emp_ID=101;

7. DQL Commands

SELECT * FROM Employee;

SELECT Emp_Name, Salary
FROM Employee;

8. WHERE Clause

SELECT *
FROM Employee
WHERE Salary > 50000;

9. ORDER BY

SELECT *
FROM Employee
ORDER BY Salary DESC;

10. GROUP BY

SELECT Department, AVG(Salary)
FROM Employee
GROUP BY Department;

11. Aggregate Functions

COUNT(), SUM(), AVG(), MIN(), MAX()

Example:
SELECT AVG(Salary)
FROM Employee;

12. JOINS

INNER JOIN:
SELECT *
FROM Employee E
INNER JOIN Department D
ON E.Dept_ID=D.Dept_ID;

LEFT JOIN, RIGHT JOIN and FULL JOIN are also available.

13. Constraints

PRIMARY KEY
FOREIGN KEY
NOT NULL
UNIQUE
CHECK
DEFAULT

14. Views

CREATE VIEW EmpView AS
SELECT Emp_Name, Salary
FROM Employee;

15. Index

CREATE INDEX idx_name
ON Employee(Emp_Name);

16. Transactions

COMMIT
ROLLBACK
SAVEPOINT